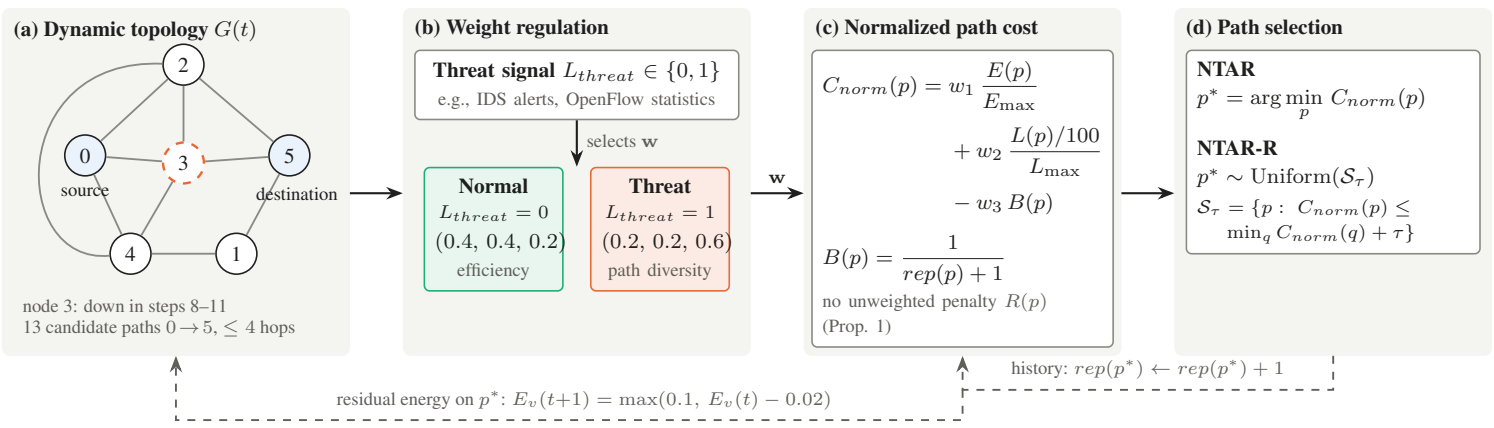

(a) Dynamic topology $G(t)$
2
0
3
5
source
destination
4
1
node 3: down in steps 8–11
13 candidate paths $0 \to 5$, ≤ 4 hops
(b) Weight regulation
Threat signal $L_{threat} \in \{0, 1\}$
e.g., IDS alerts, OpenFlow statistics
selects $\mathbf{w}$
Normal
$L_{threat} = 0$
(0.4, 0.4, 0.2)
efficiency
Threat
$L_{threat} = 1$
(0.2, 0.2, 0.6)
path diversity
$\mathbf{w}$
(c) Normalized path cost
$C_{norm}(p) = w_1 \frac{E(p)}{E_{\max}} + w_2 \frac{L(p)/100}{L_{\max}} - w_3 B(p)$
$B(p) = \frac{1}{rep(p)+1}$
no unweighted penalty $R(p)$
(Prop. 1)
(d) Path selection
NTAR
$p^* = \arg\min_p C_{norm}(p)$
NTAR-R
$p^* \sim \text{Uniform}(\mathcal{S}_\tau)$
$\mathcal{S}_\tau = \{p : C_{norm}(p) \leq \min_q C_{norm}(q) + \tau\}$
history: $rep(p^*) \leftarrow rep(p^*) + 1$
residual energy on p^*: $E_v(t+1) = \max(0.1,\ E_v(t) - 0.02)$